DUAL ROBOTIC ARM AUTONOMOUS CHESS SYSTEM

Project Proposal

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Contents

1. Project Overview
2. Project Concept
3. User Interaction
4. Main Objectives
5. Overall System Architecture
6. Robotic Arm Mechanical Design
7. Actuation System
8. Kinematic Modeling
9. Workspace Analysis
10. Trajectory Generation
11. Control System
12. ROS 2 and Robot Modeling
13. Simulation
14. Computer Vision
15. Chess Engine
16. Chess Piece Manipulation
17. Two-Robot Coordination
18. Mobile Application
19. Development and Execution Plan
20. Team Distribution
21. Risk Management
22. Minimum Viable System
23. Expected Final System

DOCUMENT NOTE	The proposal is organized for compact academic reading. Sections continue naturally across pages; page breaks are used only where they improve structure and readability.

1. Project Overview

The project aims to design, simulate, control, and build two custom robotic arms capable of playing chess on a shared chessboard.

The system will combine:

- Mechanical design and CAD
- Robotic arm design and fabrication
- Stepper motor actuation
- Kinematic modeling
- Inverse and forward kinematics
- Workspace analysis
- Trajectory generation
- PID-based control
- Computer vision
- ROS 2
- Gazebo and RViz simulation
- Chess engine integration
- Custom chess-piece gripper
- Two-robot coordination
- Mobile application interface
- Physical prototype integration and testing

The two robotic arms will use the same mechanical design with mirrored configurations. This allows the team to reuse the mechanical design, motor selection, kinematic model, URDF, control architecture, and software components while reducing development time and complexity.



Figure 1. Dual robotic-arm chess concept reference.

2. Project Concept

The main concept is to create a robot-vs-robot autonomous chess system.

Each robotic arm represents one player and operates on its side of the chessboard.

The system will be able to:

1. Determine the current chessboard state.
2. Identify the required chess move.
3. Convert the chess move into physical board coordinates.
4. Calculate the required robot joint positions using inverse kinematics.
5. Generate a smooth trajectory.
6. Control the robot joints.
7. Move the gripper to the chess piece.
8. Pick up the piece.
9. Move it to the target square.
10. Release the piece.
11. Handle captures when required.
12. Update the board state.
13. Continue until the game ends.

3. User Interaction

The system will also support a high-level user interaction mode through a mobile application.

The application will not directly control individual joints or motors.

Instead, the user can specify a chess move such as:

Knight: G1 → F3

Chess Move → Board Coordinates → Robot Coordinates → IK → Trajectory → Controller → Gripper → Physical Move

This keeps the application simple while allowing the robotic system to handle the actual motion planning and control.

4. Main Objectives

The project aims to achieve the following:

- Design two custom robotic arms.
- Use identical/symmetrical arm architectures.
- Determine the required DOF and joint configuration.
- Define link dimensions and joint limits.
- Analyze the required workspace.
- Select suitable stepper motors.
- Design motor mounts, shafts, bearings, and transmissions.
- Design a custom chess-piece gripper.
- Perform torque and motor calculations.

- Develop the robot kinematic model.
- Implement inverse kinematics.
- Use forward kinematics for validation and workspace analysis.
- Generate smooth trajectories.
- Implement PID-based joint control.
- Build ROS 2 robot models using URDF/Xacro.
- Simulate the arms in Gazebo.
- Visualize and debug the system using RViz.
- Develop a controlled computer vision system.
- Integrate an open-source chess engine.
- Implement chess-piece pickup and placement.
- Coordinate the two robotic arms safely.
- Develop a high-level mobile application.
- Integrate the complete system.
- Manufacture and assemble the physical prototype.
- Validate the physical robot against the simulation.

5. Overall System Architecture

The system will be divided into several main layers.

Input Layer	Camera Mobile application Chessboard
Perception and Game Layer	Board detection Board calibration Chess-piece detection Board-state estimation Chess engine
Robotics Layer	Chess move to board coordinates Board coordinates to robot coordinates Inverse kinematics Trajectory generation Motion control Gripper control
Robot Layer	Robot Arm 1 Robot Arm 2 Stepper motors Motor drivers Encoders if implemented Custom grippers
Simulation Layer	URDF/Xacro ROS 2 Gazebo RViz
User Interface Layer	Mobile application High-level move commands System status Start/stop controls

6. Robotic Arm Mechanical Design

The two robotic arms will be designed as identical mechanisms with mirrored coordinate frames.

The mechanical design will include:

- DOF selection
- Joint configuration
- Link dimensions
- Joint limits
- Base dimensions
- Motor mounting
- Shafts
- Bearings
- Mechanical transmission

- Structural components
- End-effector mounting
- Chess-piece gripper

The arm dimensions will be selected based on the chessboard size and required workspace rather than designing an unnecessarily large robot.

The design will prioritize:

- Sufficient workspace
- Low mass
- Mechanical rigidity
- Easy manufacturing
- Low cost
- Simple assembly
- Accurate positioning
- Safe operation



Figure 2. Mechanical components, gripper, and precision-part reference.

7. Actuation System

The project will use stepper motors as the main actuators.

Motor selection will consider:

- Required joint torque
- Arm/link mass
- Chess-piece payload
- Required acceleration
- Required operating speed
- Transmission ratio
- Holding torque
- Safety factor

The team will determine whether direct drive or mechanical transmission is more appropriate for each joint.

If feasible, encoders may be added to provide position feedback and improve the reliability of the control system.

8. Kinematic Modeling

A complete kinematic model will be developed for the robotic arm.

Inverse Kinematics

Inverse kinematics will be the main method used during operation.

Desired End-Effector Position → Required Joint Angles

For example:

Chess Square E4 → Robot Coordinates → Joint Angles

The IK solution will then be passed to the trajectory-generation system.

Forward Kinematics

Forward kinematics will mainly be used for:

- Validating IK results
- Workspace analysis
- Checking the position of the end-effector
- Simulation verification

The system will therefore use:

IK for motion generation

and

FK for validation and analysis.

9. Workspace Analysis

The required workspace will be analyzed before finalizing the arm dimensions.

The analysis will consider:

- Chessboard dimensions
- Board location relative to the robot base
- Link lengths
- Joint limits
- End-effector orientation
- Reachability of all chess squares
- Possible singular configurations
- Mechanical interference
- Collision boundaries

The final arm geometry will be selected so that the robot can reliably reach all required chessboard positions.

10. Trajectory Generation

The IK system provides target joint positions, but the robot should not instantly jump between positions.

Therefore, a trajectory-generation layer will be implemented.

The trajectory generator will produce smooth:

- Position

- Velocity
- Acceleration

profiles.

Possible approaches include:

- Polynomial interpolation
- Trapezoidal velocity profiles

The trajectory will respect the robot's velocity and acceleration limits.

This will provide smoother motion, reduce vibration, and improve the safety of chess-piece manipulation.

11. Control System

A PID-based control system will be used as the baseline controller.

The controller will receive the desired joint trajectory and generate the required actuator commands.

The control architecture will be developed around:

Desired Position → Trajectory → Controller → Motor

If encoder feedback is implemented, the system will become a closed-loop position-control system:

Desired Position → PID → Motor → Encoder Feedback

More advanced control methods such as MPC may be investigated as an optional extension if the core system is completed early.

12. ROS 2 and Robot Modeling

The robotic arms will be integrated into ROS 2.

The development pipeline will be:

CAD → Mesh → URDF/Xacro → ROS 2 → Gazebo → RViz

The URDF/Xacro model will contain:

- Links
- Joints
- Joint limits
- Visual properties
- Collision geometry
- Mass
- Inertia
- Coordinate frames

Two instances of the robot will then be created in the simulation environment.

13. Simulation

The project will include a complete simulation environment before final physical integration.

Gazebo

Gazebo will be used for:

- Robot physics
- Joint movement

- Collision detection
- Motion testing
- Chess-piece manipulation
- Two-robot interaction

RViz

RViz will be used for:

- TF visualization
- Robot model visualization
- Coordinate-frame debugging
- Sensor visualization
- ROS 2 system debugging

The simulation will be used to test the robotics pipeline before applying it to the physical prototype.

14. Computer Vision

Computer vision will be used to determine the chessboard state.

To reduce unnecessary complexity, the initial system will use a controlled environment:

- Fixed camera position
- Known chessboard dimensions
- Controlled lighting
- Known board orientation

The vision pipeline will include:

Camera → Board Detection → Calibration → Square Detection → Piece Detection → Board State

The system does not need to solve a completely general chess-recognition problem. The goal is reliable recognition within the controlled project environment.

15. Chess Engine

An open-source chess engine such as Stockfish will be integrated into the system.

The team will not develop a chess AI from scratch.

The chess engine will handle:

- Board-state processing
- Legal move generation
- Move selection
- Game progression

The robotics system will focus on converting the selected chess move into a physical robot action.

The overall flow will be:

Board State → Chess Engine → Chess Move → Robot Motion

16. Chess Piece Manipulation

A custom end-effector will be designed specifically for the chess pieces.

A simple parallel gripper will be considered as the primary design.

The gripper must reliably:

- Approach the piece
- Grip the piece
- Lift it
- Move it
- Release it accurately

The design will also consider captured pieces and situations where a target square is already occupied.

The gripper will be developed alongside the robotic arm rather than being treated as a separate final-stage component.

17. Two-Robot Coordination

Both robotic arms will operate on the same chessboard.

The initial system will use sequential operation to reduce coordination complexity.

For example:

Robot 1 moves → Robot 1 finishes → Robot 2 moves

The coordination system will ensure:

- Safe robot positions
- No arm-to-arm collision
- No unnecessary simultaneous movement
- Correct player turn
- Correct chessboard state
- Safe gripper operation

More advanced simultaneous motion can be considered as an optional extension after the core system is stable.

18. Mobile Application

A mobile application will provide a high-level interface for interacting with the system.

The application can include:

- Selecting the robot/player
- Selecting source square
- Selecting destination square
- Sending a chess move
- Starting/stopping the system
- Selecting operating mode
- Displaying system status

Example:

Knight → G1 → F3

The application will not directly control:

- Joint angles
- Motor steps
- Individual motors

Instead, the robotic system will automatically calculate and execute the required movement.

The final communication technology may be selected from:

- Wi-Fi
- Bluetooth/BLE
- Ethernet
- USB

The exact communication method will be finalized according to the hardware architecture.

19. Development and Execution Plan

Instead of dividing the project into fixed calendar weeks, development will be organized into milestones that continue until the complete prototype is assembled and operational.

Phase 1 — Requirements and System Definition

Define the complete system requirements.

Determine chessboard dimensions.

Define robot workspace.

Determine required DOF.

Define initial robot architecture.

Define system architecture.

Define camera requirements.

Define gripper requirements.

Define motor requirements.

Define ROS 2 architecture.

Divide responsibilities among team members.

Phase 2 — Mechanical Concept and Initial CAD

Develop the initial robotic-arm concept.

Select joint configuration.

Determine link dimensions.

Define joint limits.

Develop the initial gripper concept.

Estimate required motor torque.

Analyze workspace.

Begin SolidWorks modeling.

Develop the initial mechanical structure.

The two arms will be based on the same design.

Phase 3 — Kinematics and Robotics Development

Develop the mathematical kinematic model.

Implement forward kinematics.

Implement inverse kinematics.

Validate IK using FK.

Analyze reachable workspace.

Define robot coordinate frames.

Convert chessboard coordinates into robot coordinates.

Test reaching all chessboard squares.

Mechanical design and kinematic development will progress in parallel.

Phase 4 — Simulation Model Development

Export CAD meshes.

Build URDF/Xacro.

Define links and joints.

Add mass and inertia.

Add joint limits.

Add collision models.

Configure ROS 2.

Spawn the robot in Gazebo.

Visualize the robot in RViz.

Create two symmetric robot instances.

Add the chessboard and chess pieces.

Phase 5 — Motion Planning and Control

Implement trajectory generation.

Define velocity and acceleration limits.

Implement PID control.

Integrate stepper motor commands.

Add encoder feedback if selected.

Test individual joints.

Test complete arm trajectories.

Test pickup and placement motions.

Phase 6 — Vision and Chess System

Configure the camera.

Calibrate the chessboard.

Detect board geometry.

Detect chess pieces.

Generate the board state.

Integrate the chess engine.

Convert engine moves into robot actions.

Test chess moves independently from the physical hardware.

Phase 7 — Full Simulation Integration

Integrate the complete software pipeline: Camera → Vision → Board State → Chess Engine → Chess Move → Board Coordinates → Robot Coordinates → IK → Trajectory → Controller → Robot → Gripper

Both robotic arms will then be tested inside the simulated environment.

The main goal of this stage is to have the complete chess-playing workflow working in simulation before final physical integration.

Phase 8 — Physical Prototype Manufacturing

After the mechanical design and simulation have reached a stable state:

Finalize CAD.

Select materials.

Manufacture mechanical parts.

Prepare shafts and bearings.

Manufacture motor mounts.

Manufacture the gripper.

Assemble the robotic arms.

Install stepper motors.

Install motor drivers.

Install encoders if used.

Prepare the electronics and power system.

Phase 9 — Mechanical Validation — Optional

If time and resources allow, additional mechanical validation may be performed using engineering analysis tools.

Possible analysis can include:

- Stress
- Deformation
- Safety factor
- Critical joints
- Motor mounts
- Main links
- Gripper structure

This analysis is considered an optional enhancement, not a requirement for the core prototype.

The main priority remains achieving a functional mechanical prototype and complete robotic system.

Phase 10 — Hardware and Software Integration

Integrate:

- Robotic arms
- Stepper motors
- Motor drivers
- Encoders if used
- Raspberry Pi/PC
- Camera
- Grippers
- ROS 2

- Controllers
- Vision
- Chess engine

The simulation and physical robot will be compared to identify differences in:

- Position
- Speed
- Timing
- Motor behavior
- Gripper accuracy
- Coordinate calibration.

Phase 11 — Prototype Calibration and Testing

Perform complete calibration of:

- Robot coordinate frames
- Joint zero positions
- Chessboard position
- Camera
- Gripper
- Motor steps
- Joint limits

Testing will progress from simple to complete:

Single Joint → Single Arm → Pick/Place → Chess Move → Two Arms → Complete Game

Phase 12 — Final Prototype Integration

The final stage will focus on closing the complete prototype.

The objective is to have:

- Two assembled robotic arms
- Functional grippers
- Working stepper-motor system
- Calibrated chessboard
- Functional camera
- Working vision system
- Chess engine integration
- IK
- Trajectory generation
- Control system
- ROS 2 communication
- Gazebo/RViz simulation
- Two-robot coordination
- Mobile application
- Complete robot-vs-robot operation

The project will be considered complete when the complete system can execute chess moves reliably from the software command/vision layer to the physical robotic arms.

20. Team Distribution

The project can be divided among the 10 team members as follows:

Mechanical Design — 3 Members	Robotic arm CAD Link dimensions Joint design Motor mounting Bearings and shafts Gripper Manufacturing Mechanical assembly
Robotics and Control — 3 Members	Kinematics IK FK Workspace analysis Trajectory generation PID Stepper control Encoder integration ROS 2 robot control Gazebo simulation
Vision and Chess System — 2 Members	Camera Calibration Board detection Piece detection Board-state estimation Chess engine Chess move processing
Integration and Application — 2 Members	ROS 2 integration Communication between modules Two-robot coordination Mobile application System-level testing Final integration

All teams will coordinate during the integration stages rather than working as completely isolated groups.

21. Risk Management

Area	Risk	Mitigation
Computer Vision	Risk: Incorrect piece or board detection.	Mitigation: Fixed camera, controlled lighting, known board dimensions, and controlled environment.
Gripper	Risk: Pieces may slip or be incorrectly positioned.	Mitigation: Design and test the gripper early using actual chess-piece dimensions.
Inverse Kinematics	Risk: Unreachable or unstable configurations.	Mitigation: Perform workspace analysis before finalizing arm dimensions and validate IK using FK.
Robot Collision	Risk: The two arms may interfere with each other.	Mitigation: Use sequential operation and predefined safe positions.
Stepper Motors	Risk: Missed steps or insufficient torque.	Mitigation: Calculate torque requirements, use an appropriate safety factor, and consider encoder feedback.
Manufacturing	Risk: Mechanical fabrication delays.	Mitigation: Complete CAD and simulation before manufacturing and identify critical components early.
System Integration	Risk: Individual modules work but fail when combined.	Mitigation: Integrate progressively rather than waiting until the final stage.
Project Scope	Risk: Too many advanced features delay the core prototype.	Mitigation: Prioritize the complete robot-vs-robot chess system first. Advanced features such as MPC, human-vs-robot operation, simultaneous two-arm motion, and additional engineering analysis remain optional extensions.

22. Minimum Viable System

The minimum successful version of the project will be:

> **Two custom robotic arms + chessboard + grippers + stepper motors + IK + trajectory generation + control + ROS 2 + Gazebo/RViz + chess engine + automated chess-piece manipulation.**

The system should first be proven in simulation and then transferred to the physical prototype.

Vision and the mobile application can be integrated progressively without blocking the core robotic system.

23. Expected Final System

The final prototype will consist of:

- Two custom-designed robotic arms
- Identical/symmetrical mechanical architecture
- Stepper motor actuation
- Custom chess-piece grippers
- Defined workspace and joint limits
- Kinematic model
- Inverse kinematics
- Forward-kinematics validation
- Trajectory generation
- PID-based control
- ROS 2 integration
- URDF/Xacro models
- Gazebo simulation
- RViz visualization
- Computer vision
- Chess engine
- Two-robot coordination
- Mobile high-level interface
- Physical prototype
- Complete robot-vs-robot chess operation

Additional engineering analysis and advanced control techniques may be added if time and resources allow, but they will not be dependencies of the core prototype.

> **The main objective is to deliver a fully integrated physical dual-arm robotic chess system, from mechanical design and simulation to control, perception, and real-world operation.**

END OF PROJECT PROPOSAL